(Please write your Enrollment Number) Enrollment No. 00601012020

MID-TERM EXAMINATION
(Course Name B.Tech,CSE/IT) (Semester 6th)
(February, 2023) OFF LINE mode

Subject Code: BCS 306	Subject: Compiler Design
Time : 1 ½Hours	Maximum Marks : 30
Note: Q. 1 is compulsory.	

Q1 **(2.5*4)**

(a) Define FIRST and FOLLOW functions, explain with an example Grammar.

(b) Define CFG? Write about Derivation trees and Parse tree.

(c) What is left recursion? Eliminate left recursion in the following grammar:

$A \rightarrow ABd \mid Aa \mid a$

$B \rightarrow Be \mid b$

(d) Define Ambiguity? Check whether the grammar S→aAB, A→bC/cd, C→cd, B→c/d, is Ambiguous or not?

Q2 **(Attempt any Two Parts)** **(5,5)**

(a) With an example explain about various phases of a compiler with a neat diagram?

(b) Discuss about bootstrapping with a neat diagram?

(c) Write about the following:
(i) Role of Lexical Analyzer & Specification of Tokens using Regular Expressions
(ii) Input Buffering

Q3 **(Attempt any Two Parts)** **(5,5)**

(a) Write about shift reduce parser and parse the input string int id,id; using shift-reduce parser for the grammar

S→TL;
T→int | float
L→L,id | id

(b) Construct LR(0) items and SLR Parsing table for the grammar S→L=R/R, L→*R/id, R→L

(c) What is the difference between Top-down Parsing and Bottom up Parsing? Write about Operator precedence parser with an example.

S int id, id; shift
S int id, id; reduce T→int
S T id, id; shift
S T L , id; shift
S T L, id ; red L→id
S T L ; shift
S T L; reduce S→TL;
S S